

Snapdragon® AI Lab Build & Present Challenge · Qualcomm

VoxShield Edge

Real-Time, Privacy-Preserving Voice-Clone Defense for Indian Telephony

A bank fraud analyst's HP Snapdragon laptop detects AI-cloned voices in real time, fully on-device on the Hexagon NPU — no call audio ever leaves the machine.

5.9%

EER · deployed fusion
(held-out In-the-Wild)

6.8%

EER · unseen-generator
deepfakes (OOD, MLAAD)

0.98

ROC-AUC
on OOD deepfakes

45–80

TOPS Hexagon NPU
(Snapdragon X / X2)

Devansh Goenka · on-device · private · Snapdragon NPU

THE PROBLEM

The problem: cloned voices over the bank phone line

AI voice cloning is cheap and fast. In India, phone-based fraud — fake bank officials, “digital arrest” scams, relative-in-distress calls — increasingly uses synthetic voices in Indian languages over ordinary 8 kHz phone lines.

- Cloning a voice now takes seconds of reference audio.
- Telephony (G.711, 8 kHz) hides the very artifacts most detectors rely on.
- Sensitive customer call audio should not be shipped to a foreign cloud.
- A verdict after the call ends is forensics, not protection.

MOTIVATION

Why on-device — and why Snapdragon, now

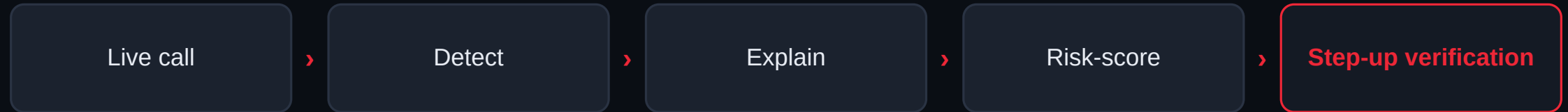
- Privacy by design: raw call audio never leaves the analyst's laptop (DPDP-aligned).
- Real-time: a verdict forms within the first seconds of the call, offline.
- The Snapdragon Hexagon NPU (45–80 TOPS) makes local real-time speech AI practical on a laptop.
- Resilient: works air-gapped, in branches with poor connectivity.

SOLUTION

VoxShield Edge

An on-device, Indic-first, telephony-hardened voice-deepfake detector. Given a live call it returns a calibrated LOW / MEDIUM / HIGH risk verdict with explainable reason codes — decision support for a fraud analyst, never automatic blocking.

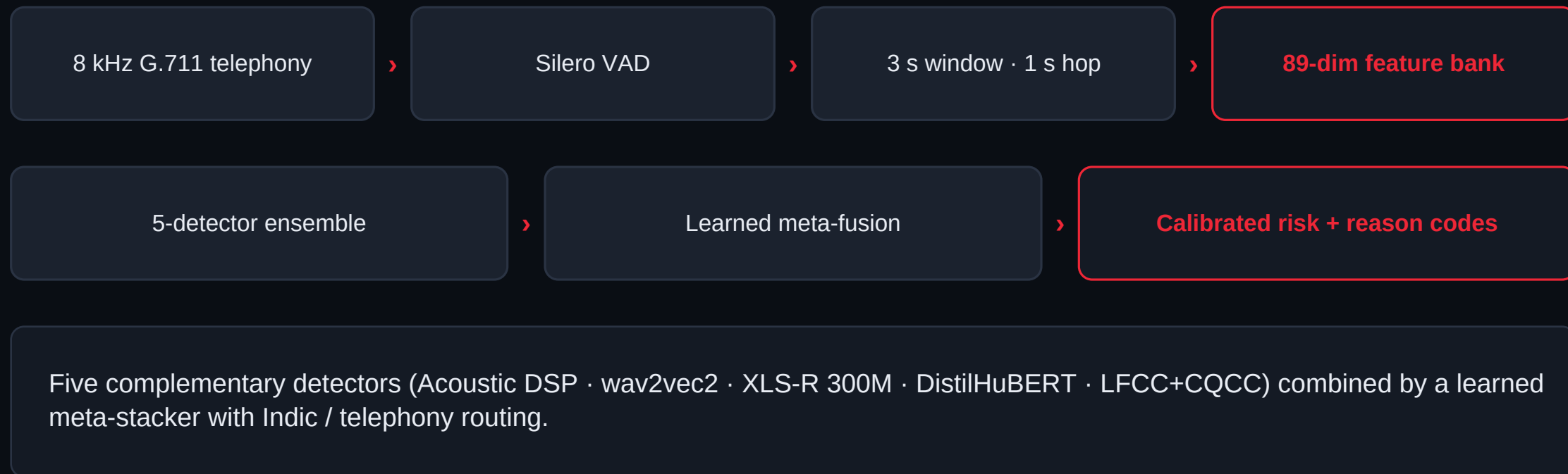
OPERATIONAL FLOW



Decision support for a fraud analyst — **never automatic blocking.**

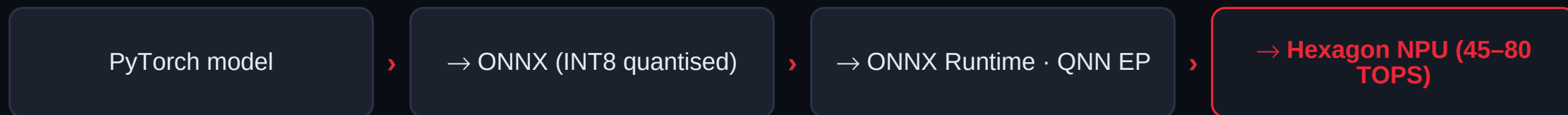
ARCHITECTURE

How it works



Snapdragon optimisation — the on-device pipeline

ON-DEVICE INFERENCE PATH



- Compiled & profiled on real Snapdragon devices via Qualcomm AI Hub (qai-hub).
- On-device Windows/ARM64 app through ONNX Runtime QNN Execution Provider (onnxruntime-qnn).
- Leverages the AI Hub speech ecosystem: WavLM-Base-Plus backbone + Whisper for on-device transcription.
- Honest status: detector runs today on CPU/GPU; NPU port + INT8 quantisation via AI Hub is the optimisation in progress.

EVIDENCE

Evidence — measured, honestly scoped

Model / setting	Evaluation	Metric	Result
Deployed fusion	held-out In-the-Wild	EER	5.9%
Unseen-generator deepfakes	MLAAD (OOD)	EER / AUC	6.8% / 0.98
Detector scoreboard	same eval set	EER single/avg/fusion	16.0 / 19.1 / 5.9
Indic recall	MMS-TTS fakes (after retrain)	recall	42% → 82%

EER = Equal Error Rate (lower is better) — not accuracy. In-corpus and out-of-distribution figures are reported separately and never conflated.

DEPLOYMENT

Deployment & accessibility

- Runs on an HP Snapdragon laptop — offline, private, low-power on the NPU.
- No cloud dependency; no raw audio egress — architected for institutional data-governance.
- Explainable analyst console: risk + per-signal reason codes (spectral, phase, prosody, breath).
- Human-in-the-loop: recommends OTP / biometric / fraud-desk step-up; never auto-blocks a customer.

DIFFERENTIATION

What makes it different

Cloud incumbents

English-first, cloud-only, closed, black-box

VoxShield Edge

Indic-first · telephony-hardened · on-device NPU ·
explainable · open/measured

The contribution is the integration under one hard constraint set — Indic + narrowband + on-device + explainable + private — not a single new algorithm.

ROADMAP

Roadmap

- Now: ONNX export + INT8 quantisation; Qualcomm AI Hub compile/profile on Snapdragon X/X2.
- Next: measured on-NPU latency + power; streaming live-call demo on the laptop.
- Then: more Indic languages, larger real-world corpus, challenge-response liveness.
- Goal: a private, on-device fraud-screening layer for every bank branch.

THANK YOU

VoxShield Edge — private AI, on your device

Indic-language + telephony-quality + real-time + on-device + explainable voice-deepfake detection for financial fraud — running on the Snapdragon NPU, with honest, measured evidence.

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Devansh Goenka · GitHub: [DG10911/voxshield](https://github.com/DG10911/voxshield) · devanshgoenka03@gmail.com